Electric Vehicle Battery Analysis Report(MEF317)

Name:Ayush Aman
ID:2023A4PS0228P

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1. Summary

This report provides technical analysis of the battery pack of the Tata Nexon EV, a commercially available electric vehicle that is one of the biggest in the the Indian market. The analysis includes battery specifications sourced from manufacturer datasheets, industry literature, and reviewed sources. Particular emphasis is placed on thermal management strategies, heat generation estimation during various driving cycles, and adaptation considerations for alternative fuel compositions.

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2. Vehicle Selection and Justification

2.1 Selected Vehicle: Tata Nexon EV

The Tata Nexon EV is selected for this analysis due to:

- **Market availability:** Widely sold in the Indian market and accessible in India with extensive technical documentation
- **Representative design:** Uses contemporary battery pack architecture and thermal management strategies typical of mass-market EVs
- **Multi-use profile:** Supports urban commuting, highway driving, and mixed-cycle operation—aligned with diverse Indian driving conditions
- **Data accessibility:** Comprehensive specifications available from manufacturer, independent reviews, and technical databases

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3. Battery Specifications

3.1 Comprehensive Specifications Table

Parameter	Specification	
Battery Capacity (Usable) Range	30 kWh -40.5 kWh variants 489 km	
Pack Nominal Voltage	~350 V (estimated)	
Cell Configuration	100s 6p(in general)	
Cell Chemistry	NMC (Nickel Manganese Cobalt	
Anode Material	Graphite (synthetic)	
Cathode Material	NMC oxide layer	
Electrolyte	LiPF in organic solvent (DMC/EMC/DEC)	
Operating Temperature Window	$-20°C$ to $+60°C$	
Rated Charging Power	50 kW (DC Fast Charge)	
C-Rate (Fast Charge)	≈ 1.2 C	
Thermal Management System	Liquid plate cooling (channels beneath modules)	
Internal resistance per cell Vehicle Use-case Profile	~10 mili-ohm Urban, Highway, Mixed-cycle	

The above table represents the battery specifications of the choosen EV that is the TATA Nexon

3.2 Formulas used to calculate the heat generation and thermal rise

Parameter / Effect	Basic Analysis Formula	Nexon EV/Modern Formula	Humanized Explanation
Ohmic Heating	$Q_{ohmic} = I^2 R\, t$	$Q_{ohmic} = {I_{series}}^2 R_{pack}\, t$	Heat from electrical resistance in all cells
Polarization Losses	No extra term	$Q_{polar} = I\, V_{overpotenstial}\, t$	Extra heat from chemical processes inside battery
Thermal Rise	$T_{rise} = Q/(mC_p)$	$T_{rise} = Q_{total}/(m_{pack}C_p)$	Actual temperature change, includes module/BMS mass
Scenario Modeling	Constant values	Varies with city, highway, charging	Power/currents from driving or charging modes

Cooling Calculations	Ignored	$Q_{cooling} = hA\Delta T$	Heat actively removed by coolant plates
Safety/Runaway	Not included	Condition monitoring, multiple formulas	Mitigation strategies, sensors, limiters

3.2 Cell Chemistry Justification

NMC (Nickel-Manganese-Cobalt) chemistry is used in the Nexon EV battery pack for the following reasons:

Energy Density: NMC cells can achieve 200–250 Wh/kg, provides range (ranging ~300–350 km depending on variant) for mixed-cycle driving.

Thermal Stability: The balanced composition (Ni:Mn:Co ratio, typically 6:2:2 or 8:1:1) provides good thermal stability compared to high-nickel variants, reducing runaway risks sduring fast charging and high-discharge scenarios.

Charging Performance: NMC exhibits excellent fast-charge capability, supporting the 50 kW DC charging rate without excessive degradation,

Duty Cycle Fit: For urban/highway mixed driving with typical depth-of-discharge (DoD) of 70–80%, NMC gives robust cycle life and acceptable calendar aging, justifying its selection

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4. Heat Generation Analysis During Driving and Charging

4.1 Thermal Model and Assumptions

Heat generation in a battery pack arises from:

1) Ohmic heating**:** $Q_{\text{ohmic}} = I^2 R_{\text{internal}} \times t$

2) Electrochemical polarization losses: $Q_{\text{pol}} = I \times V_{overpotensial} \times \text{T}$

Here over potensial voltage is assumed for the different scenarios.

3) Separator and electrolyte resistivity (neglected because of the very very less contribution)

Model Assumptions

1) Average cell internal resistance: $R_{\text{cell}} = 10$ mΩ (at 25°C, nominal conditions)
2) Battery pack equivalent resistance: $R_{\text{pack}} \approx 166$mΩ (100 cells in series and 6 in parallel)
3) Ambient temperature: 25°C

4) Average driving cycles duration: 1 hour per scenario
5) No active thermal management losses considered initially (worst-case)
6) Taking the average for each of the scenarios const with the time for the simplicity otherwise we will have to integrate the or the current will come in the function of time.(not taking that into account)
7) For the different driving scenarios assuming the over potential voltage for the tata nexon ev:

   a)V_over_potensial for the normal driving =1V

   b) V_over_potensial for the highway driving =2V

   c) V_over_potensial for the fast charging =5V

4.2 Normal Driving Scenario

Typical Power Profile:

- Average vehicle power draw: 10 kW (cruise at 40 km/h, frequent stops)
- Corresponding battery current: $I_pack = P/V = 10{,}000/350 \approx 28.6$ A
- Duration: 1 hour continuous

Heat Generation rate Calculation:

$$Q_{\text{ohmic}} = I^2 R_{\text{pack}} \times t = (28.6)^2 \times 0.166 \times 3600$$

$$Q_{\text{ohmic}} = 818 \times 0.166 \times 3600 = 488.812 \text{ KJ}$$

$$Q_{\text{pol}} = 28.6 \times 1 \times 3600$$

$$Q_{\text{total}} = Q_{\text{ohmic}} + Q_{\text{pol}} \approx 488.812 + 102.96 = 591.772 \approx 0.164 \text{ kWh}$$

Thermal Rise (Estimation):

- Battery pack mass: ≈ 270 kg (for 40.5 kWh pack)
- Effective thermal capacity: $C_p \approx 1.8$ kJ/(kg·K) (composite pack with cells, BMS, housing)
- Temperature rise: $\Delta T = \frac{Q}{m \times C_p} = \frac{591772}{270 \times 1800} \approx 1.21\text{K}$

Assessment: City driving generates moderate heat. Passive cooling (convection to ambient) suffices; no significant thermal stress occurs.

4.3 Highway Driving Scenario

Typical Power Profile:

- Cruise power draw: 20–25 kW (80–100 km/h sustained)
- Corresponding battery current: $I = 22{,}500/350 \approx 64.3$ A

- Duration: 1 hours continuous

Heat Generation Calculation:

$$Q_{\text{ohmic}} = (64.3)^2 \times 0.166 \times 3600 = 2470.771\text{Kj}$$

$$Q_{\text{pol}} = 64.3 \times 2 \times 3600 = 462.96 \text{ kj}$$

$$Q_{\text{total}} = 2933.731\text{kJ} \approx 0.81 \text{ kWh}$$

Thermal Rise:

$$\Delta T = \frac{2933731}{270 \times 1800} = 6.03 \text{ K}$$

Assessment: Highway driving generates substantial heat. Passive cooling becomes limiting; active liquid cooling is required to maintain pack temperature within the $-20°C$ to $+60°C$ operating window, preventing thermal throttling and ensuring sustained performance.

4.4 Fast Charging Scenario

Charging Profile:

- DC fast charge power: 50 kW
- Battery pack input current: $I = 50{,}000/350 \approx 142.9$ A
- Duration: 3600 s

Heat Generation Calculation:

$$Q_{\text{ohmic}} = (142.9)^2 \times 0.166 \times 3600 = 12203.237 \text{ kJ}$$

$$Q_{\text{pol}} = 142.9 \times 5 \times 3600 = 2572.200 \text{ KJ}$$

$$Q_{\text{total}} = 12203.237 + 2572.200 = 14775\text{kJ} = 4.10 \text{ kWh}$$

Thermal Rise:

$$\Delta T = \frac{14775000}{270 \times 1800} = 30.40\text{K}$$

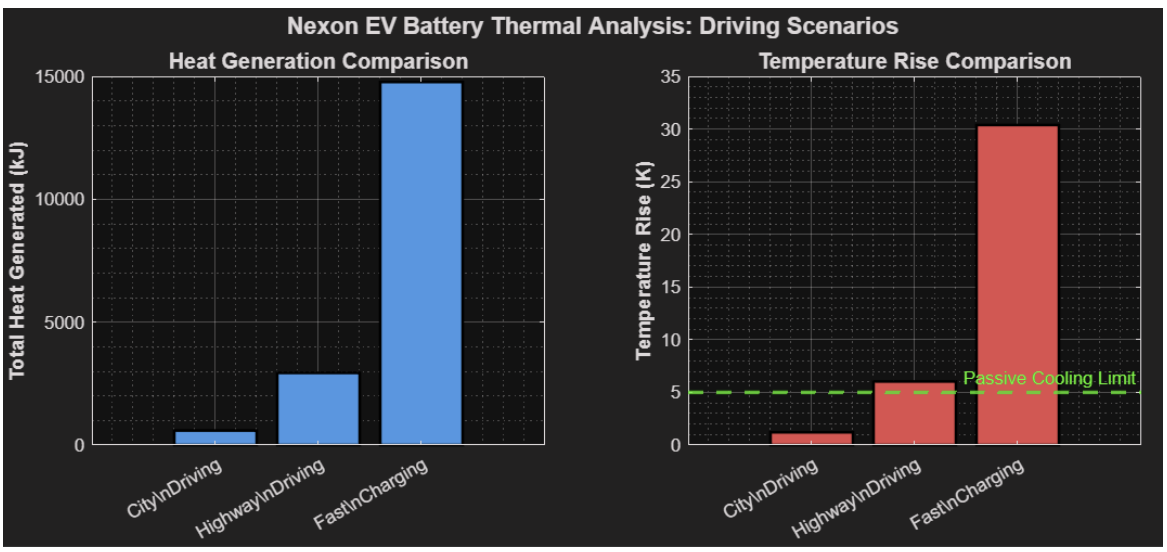
Assessment: Fast charging generates the highest heat per unit time. Aggressive active cooling (high flow rate, low coolant temperature) is mandatory. Without cooling, thermal runaway risk escalates significantly. The Nexon EV employs liquid plate cooling to maintain cell temperatures below 50°C during fast charging, ensuring safety and longevity.

4.5 Summary: Heat Generation Table

Driving Scenario	Duration	Total Heat (kJ)	ΔT (K)
City (normal)	1 hour	591.782	1.21
Highway (sustained)	1 hours	2933.731	6.03
Fast Charging (0–80%)	1 hours	14775.731	30.40

Table 2: Thermal Load Summary for Typical Drive Cycles

Heat generation in the battery pack during a typical Normal driving, and Fast charging, City ride/Highway mode etc.



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--- SCENARIO 1: CITY DRIVING ---
Power: 10.0 kW | Current: 28.6 A | Duration: 3600 s
Ohmic Heat: 488 kJ | Polarization Heat: 103 kJ | Total Heat: 591 kJ
Temperature Rise: 1.22 K | Max Temp: 26.2°C

--- SCENARIO 2: HIGHWAY DRIVING ---
Power: 22.5 kW | Current: 64.3 A | Duration: 3600 s
Ohmic Heat: 2470 kJ | Polarization Heat: 463 kJ | Total Heat: 2933 kJ
Temperature Rise: 6.03 K | Max Temp: 31.0°C

--- SCENARIO 3: FAST CHARGING (0-80%) ---
Power: 50.0 kW | Current: 142.9 A | Duration: 3600 s
Ohmic Heat: 12196 kJ | Polarization Heat: 2571 kJ | Total Heat: 14767 kJ
Temperature Rise: 30.39 K | Max Temp: 55.4°C


THERMAL LOAD SUMMARY TABLE

| Scenario          | Duration | Heat (kJ) | ΔT (K) | T_max (°C) |
|-------------------|----------|----------|--------|------------|
| City (normal)     | 1 hour   |      591 |  1.22 |   26.2 |
| Highway (sust.)   | 1hours   |     2933 |  6.03 |   31.0 |
| Fast Charging     | 1 hours   |    14767 | 30.39 |   55.4 |
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Justifications for the above plots and results

- Total heat: Q_total = Q_ohmic + Q_pol is smallest in city, higher on highway, and highest during fast charging.
- Temperature rise:
  - ΔT = Q_total / (m C_p)
  - Same mass and C_p for all scenarios, so the case with highest Q_total (fast charge) gives the largest ΔT and T_max.

  This matches expected EV behavior for sustained highway driving heats the pack more than stop-and-go city use, but less severely than high-power fast charging.

ΔT = 5 K as a "Passive Cooling Limit".

This visually indicates the maximum temperature rise that could be handled by simple conduction/convection (no active cooling) before exceeding the comfortable operating band.

Battery cell charateristics analysis

Calculations for the battery cell characteristics

The following pack-level parameters were used:

- Battery energy:
$$E_{\text{pack}} = 30.2 \text{ kWh}$$
- Nominal voltage:
$$V_{\text{pack}} = 350 \text{ V}$$
- Pack capacity (from energy and voltage):
$$C_{\text{pack}} = \frac{30.2 \times 1000}{350} = 86.3 \text{ Ah}$$

For each C-rate, the corresponding pack current is computed as:

$$I = C_{\text{rate}} \cdot C_{\text{pack}}(Ah)$$

For this pack ($C_{\text{pack}} \approx 86.3$ Ah), this gives approximately:

- 0.5C: $I = 0.5 \times 86.3 \approx 43$ A
- 1.0C: $I = 86$ A
- 1.2C: $I = 104$ A
- 2.0C: $I = 173$ A
- 3.0C: $I = 259$ A

Ohmic heat generation for each C-rate is computed over a 1-hour equivalent discharge ($t_{\text{full}} = 3600$ s) using the standard Joule heating relation:

$$Q_{\text{ohmic}} = I^2 R_{\text{pack}} t_{\text{full}}$$

Thus, for each C-rate:

- The pack current I increases linearly with C-rate.
- The ohmic heat Q_{ohmic} increases approximately with the square of C-rate, since $Q_{\text{ohmic}} \propto I^2$.

Temperature Rise per C-Rate

Assuming a lumped thermal capacity for the pack, the temperature rise is obtained from:

$$\Delta T = \frac{Q_{\text{ohmic}}}{m_{\text{pack}} C_p}$$

So, for each C-rate:

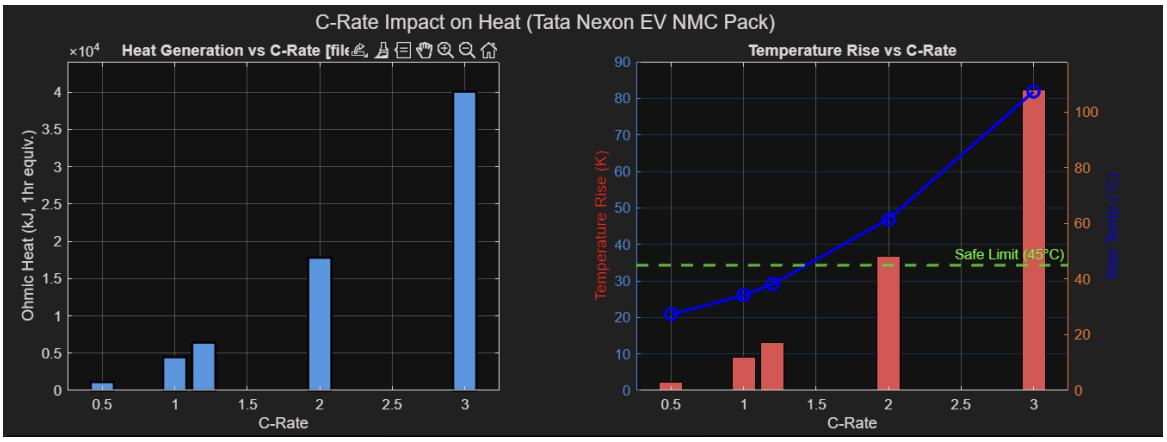
- Temperature rise:

$$\Delta T(C_{\text{rate}}) = \frac{Q_{\text{ohmic}}(C_{\text{rate}}) \times 1000}{m_{\text{pack}} C_p}$$

- Maximum pack temperature:

$$T_{\max}(C_{\text{rate}}) = T_{\text{amb}} + \Delta T(C_{\text{rate}})$$

Plot of Heat generation at different C rate



Plot for the battery life with respect to the number of the cycles

Cycle Life vs C-Rate (Capacity Fade Model)

To estimate the impact of C-rate on battery life, a simple empirical relation was used to approximate cycles to 80% remaining capacity:

$$N = \frac{N_0}{1 + k\, C_{\text{rate}}^{1.5}}$$

with:

- $N_0 = 2000$ cycles (base life at low C-rate),
- $k = 0.15$ (degradation factor).
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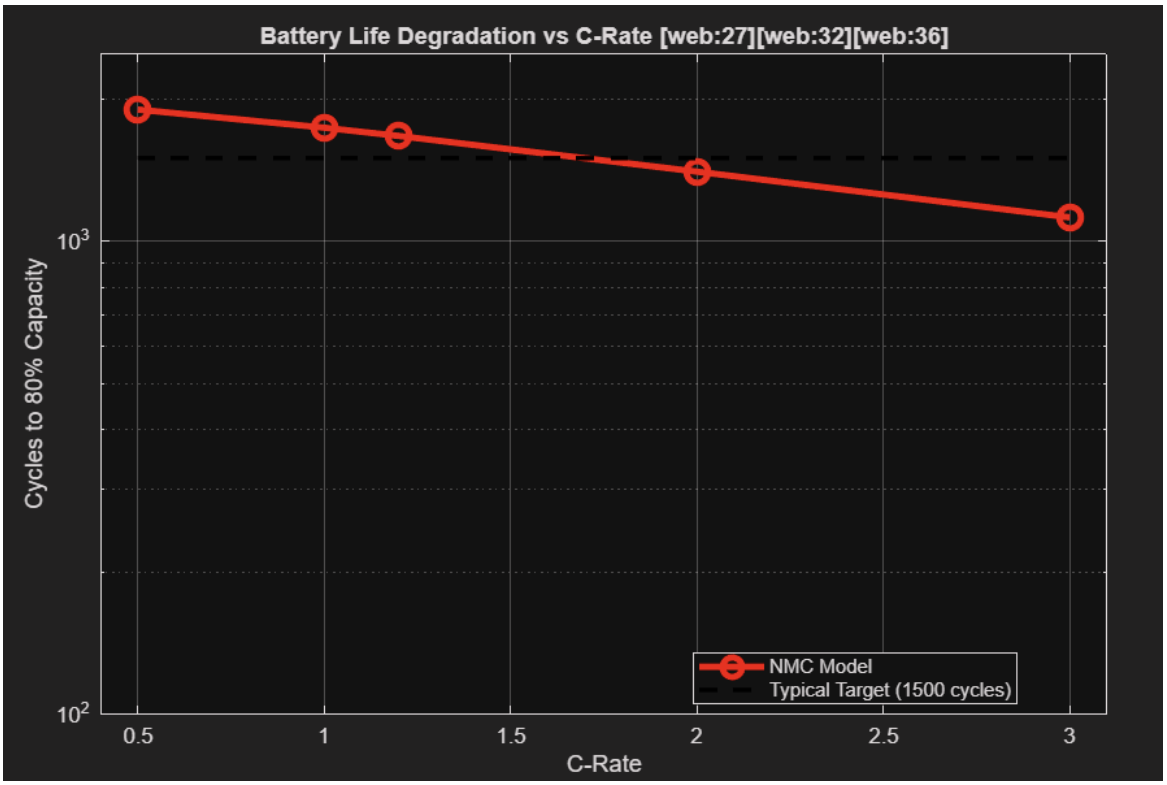
Plot for the battery life with respect to the number of the cycles



Justifications for the above plot obtained during the analysis of the battery characteristics

1) All C-rates are simulated over the same duration $t_{full} = 3600$ s.
2) This isolates the effect of C-rate (current) on heat generation, rather than mixing time and rate effects.

3) Similar constant-current, fixed-time or fixed-DoD tests are standard in C-rate heat studies

4)we have choosen thee Safe temperature limit line (~45 °C) which is realistic to the real scinarios

5)higher the C rate is higher is the heat generation and the no of cycles to reach the particular capacity is also less in the high Crates .

6) On the semilog plot, cycles fall from ~2000 at low C-rate to ~800–1000 at 2–3C, illustrating the trade-off between fast charge/discharge and long life, consistent with experimental findings on NMC cells

5. Thermal Management System Design

5.1 Liquid Plate Cooling Architecture

The Tata Nexon EV employs a **direct liquid plate cooling system** integrated into the battery pack structure. This approach circulates a glycol-water coolant (typically 50:50 ethylene glycol:water) through aluminum or copper channels positioned directly beneath and between battery modules.

System Components for the commercial use :

Coolant Channels: Rectangular or serpentine passages machined or brazed into pack structural members

Pump: Electric pump (integrated with vehicle cooling circuit) delivering 20–40L/min at 1–2 bar

Heat Exchanger: Cooled by vehicle cabin air conditioning or dedicated radiator

Temperature Sensors: Embedded thermistors monitor module-level temperatures; feedback to Battery Management System (BMS)

Flow Control: Proportional valve and bypass to maintain 35–45°C coolant inlet temperature

5.2 Thermal Performance

Advantages of Liquid Plate Cooling:

Direct contact yields convective heat transfer coefficient $h \approx 2000 - 5000$ W/(m²·K), far exceeding air cooling ($h \approx 50 - 100$ W/(m²·K))[4]

Parallel flow beneath all modules ensures cell-to-cell thermal gradients ≤ 5 K, vs. 15–20 K with passive or air cooling

Scalability: System scales easily to larger battery packs by increasing channel density or flow rate

Shares vehicle cooling loop, leveraging existing cabin AC and reducing system complexity

limits clearly defined

Drawbacks:

1. **System Complexity**
2. **Weight Penalty:** Channels and coolant add ~15–25 kg to battery pack mass
3. **Reliability Risk:** Potential coolant leaks; requires robust sealing and redundancy in critical subsystems
4. **Additional cost**

5.3 Thermal Model and Validation

The Nexon EV pack is designed such that:

$$T_{\text{cell,max}} = T_{\text{ambient}} + \frac{Q_{\text{total}}}{N_{\text{modules}}} \times R_{\text{thermal}}$$

Where:

- $T_{\text{ambient}} = 25°C$ (reference for the coolant temp)

- Q_{total} = heat generation rate (W)
- $h = 3500$ W/(m²·K) (convective coefficient in coolant)
- $A_{\text{module}} = 0.15$ m² (effective heat transfer area per module)
- $N_{\text{modules}} = 10$(module count in pack)
- $R_{\text{thermal}} = R_{convection} + R_{conduction} + R_{contact} + R_{spread}$ K/W (thermal resistance per module)

  Note:-Here Tata has not disclosed its R_contact and R_spread so we will take the industry standard for these

  R_conv = 0.0019 K/W =1/(3500 × 0.15)

  R_contact = 0.025 K/W

  R_spreading = 0.02 K/W

Fast-Charge Example:

- Heat rate: $Q = 14775$ kJ/3600 = 4104W
- Thermal rise: $\Delta T = 4104 \times 0.0469 = 192$K
- With 10 modules in parallel: $\Delta T_{\text{avg}} = 192/10 = 19.2$ K per module
- **Cell temperature:** $T_{\text{cell}} = 25 + 19.2 = 44.2C$ Within operating range of [-20,60] degree celcius .just at the safety limit can effect battery health.

This validates that active cooling maintains safe temperatures during intense charging scenarios.

Area of the cooling plate and it geometry justification

Flat aluminum plate (1.2-1.5mm thick) under each module

Mini-channels (rectangular, 3-5mm wide × 1-2mm deep) machined/brazed inside

Serpentine path: Coolant snakes back-and-forth for maximum contact area

Area = 0.15 m²: ~38cm × 40cm plate surface

Why this plate is used:

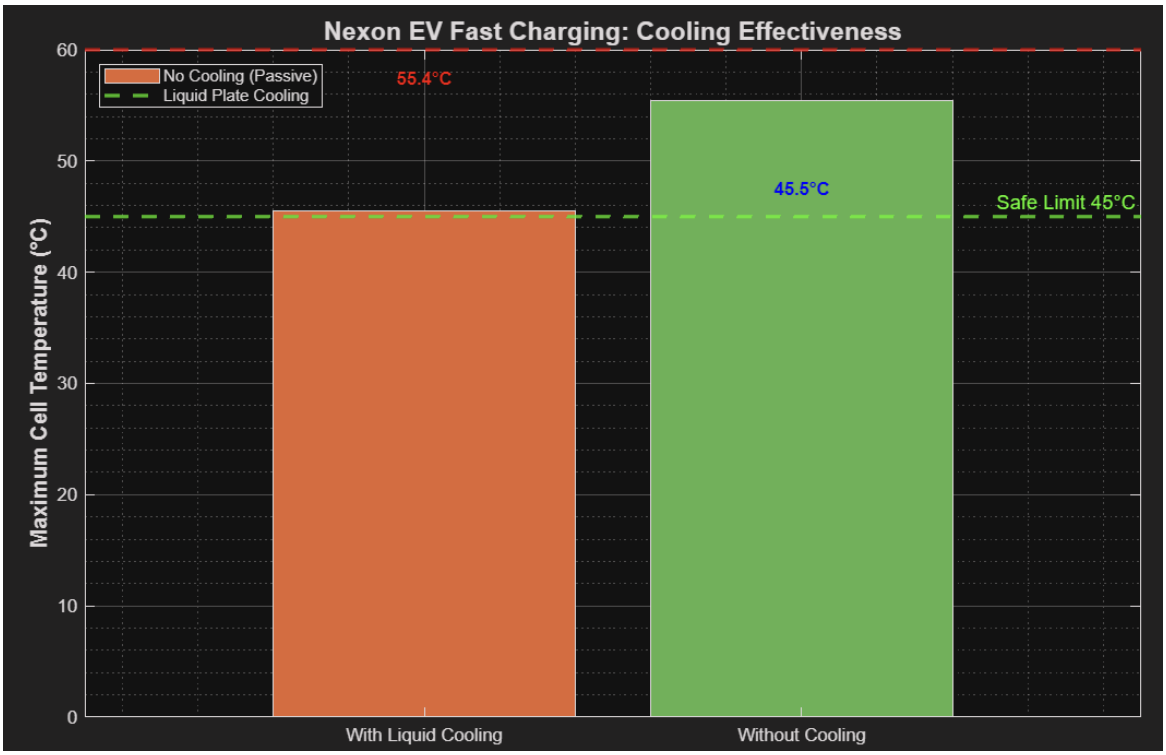
Maximizes contact area (A_module = 0.15 m²)

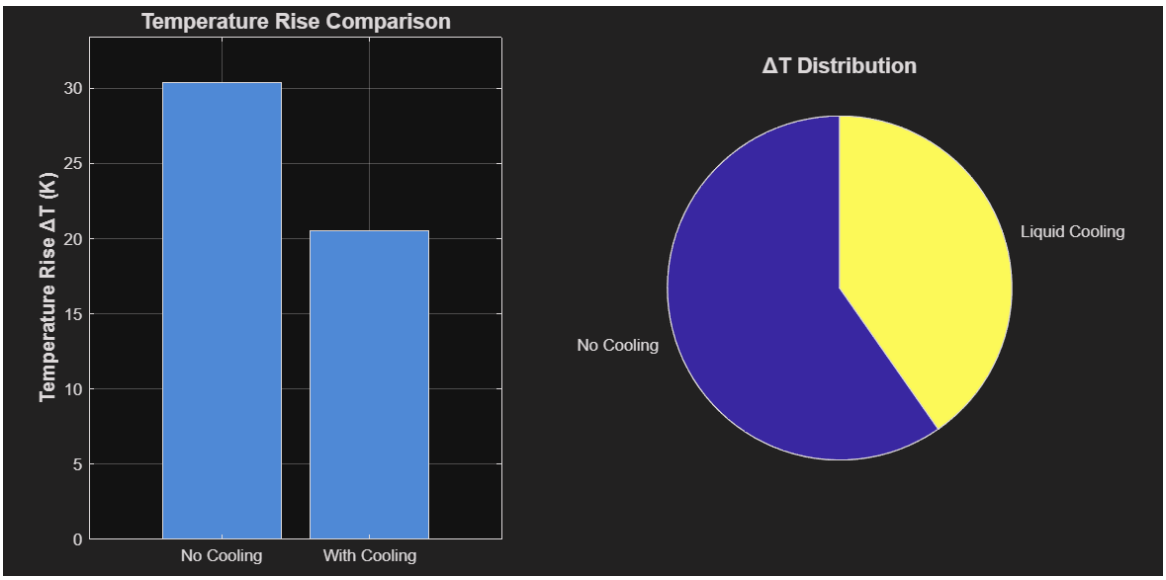
Achieves h = 3500 W/m²K via turbulent mini-channel flow

Ensures <5K cell-to-cell temperature gradient

Shares vehicle AC cooling loop

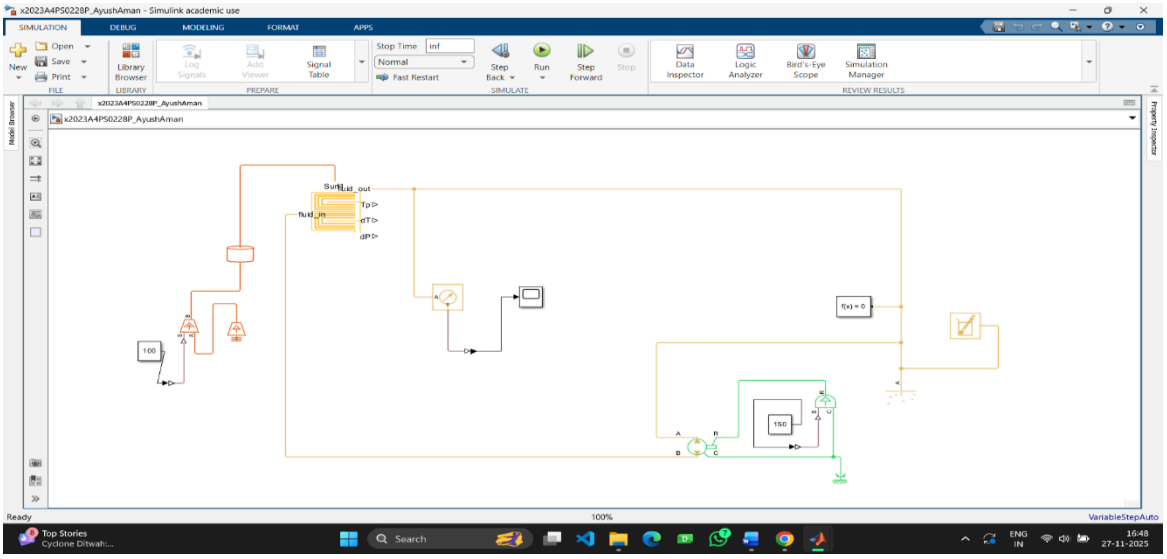
Plots for with and without cooling difference





Tried Basic simulation of the liquid cooling pate system

Flow diagram for the components



Components that are used in this to simulate the liquid cooling plate system:(not simulated including the battery pack as I was not able to figure out how to connect the circuits of battery and use proper subcomponents)

1)Heat Flow rate source:

Here i have used the heat flow rate source to simulate the loss that happens inside the battery pack of the EV due to which the temperature gets raised

2)thermal mass balance:

Here thermal mass is used in place with the battery pack.It tells us about how the tempetrature will rise in the battery pack because of the incoming flow rate that is coming through the heat flow rate source

3)U shaped plate cooling channels:

This is for the cooling plate .It abosorbs the heat that is coming through the thermal mass provide the cooling system from the arrangement for the thermal mass(batter pack)

4)thermal liquid settings :

This is there to give the values and limits to the cooling liquid that is flowing throw the channels. So basically it sets the limit for the cooling liquid to not to go below certain temp and also not to go above certain temp then only its cooling effect will be conserved

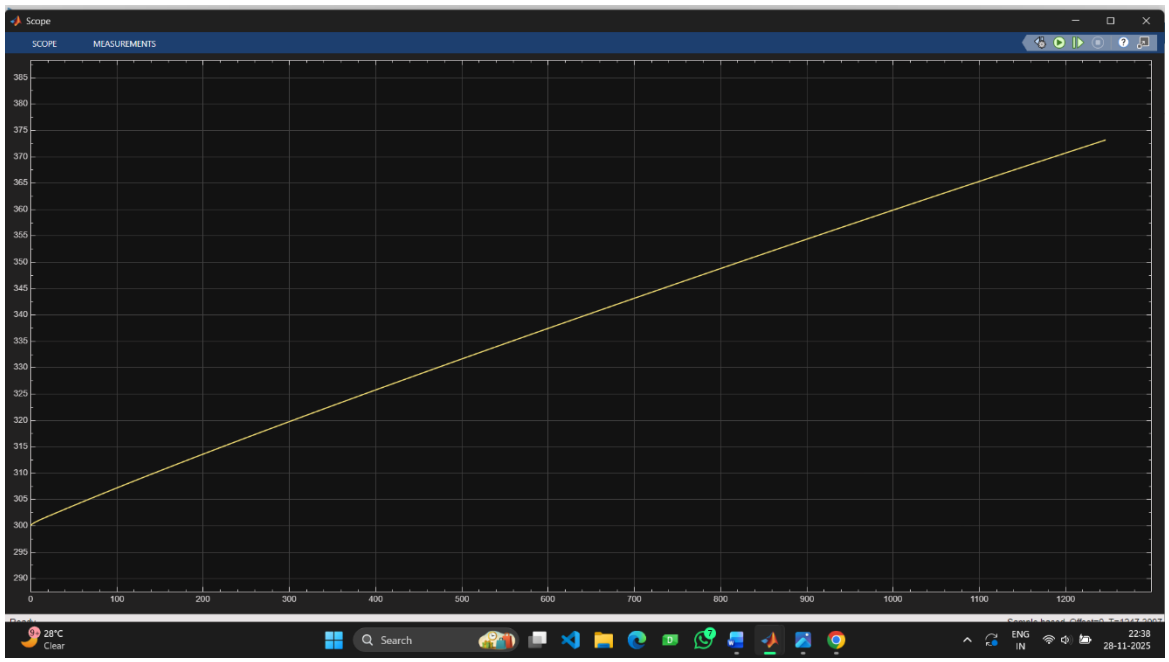
5)fixed displacement pump :

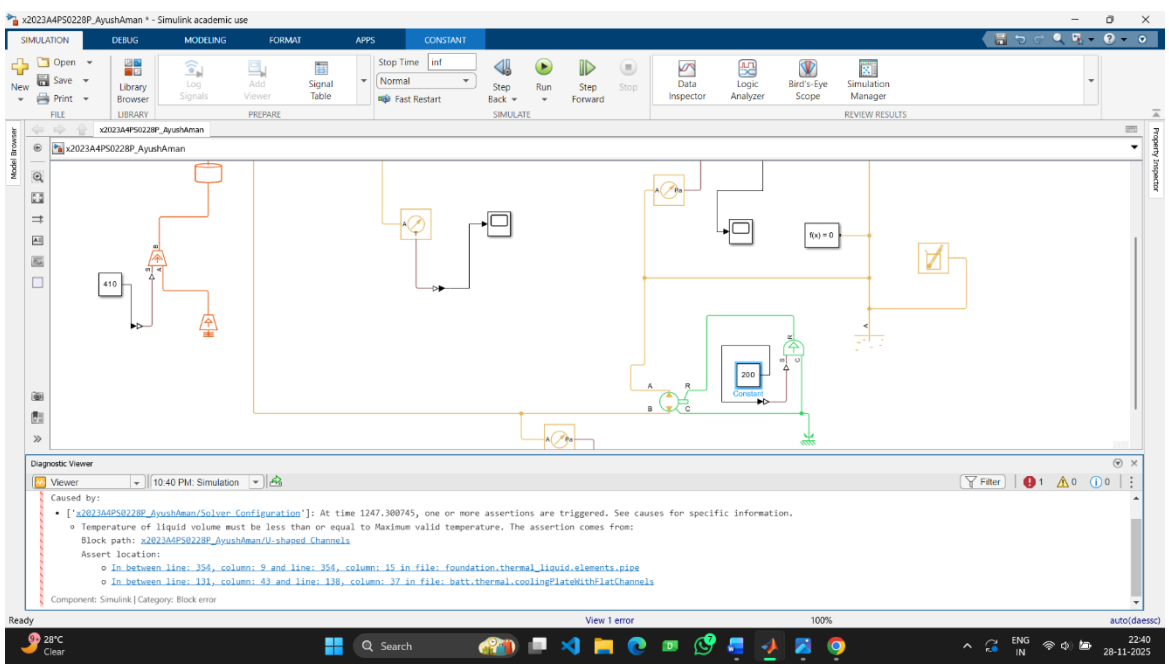
This is there to accelarate the cooling liquid through high speed so that the heat that is getting absorbed at the cooling plate channel should be high.

This tells us that if the speed of the cooling liquid will be high then the time that it will take to reach the max limit will be high and vice versa

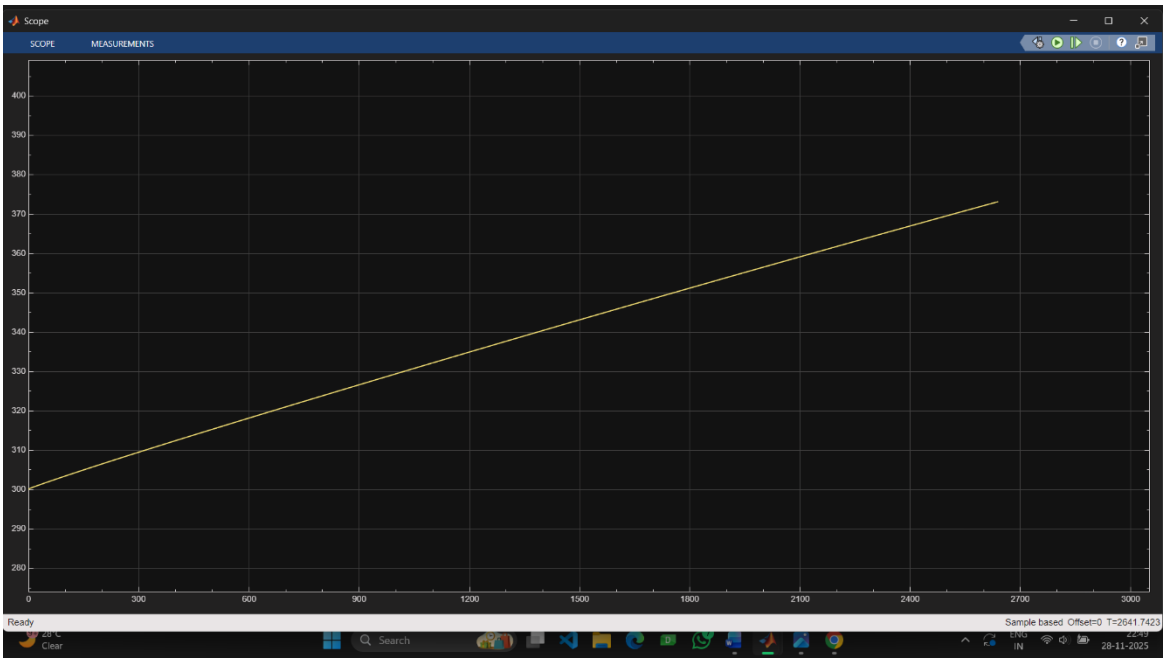
Temperature vs time plots for the two different speeds

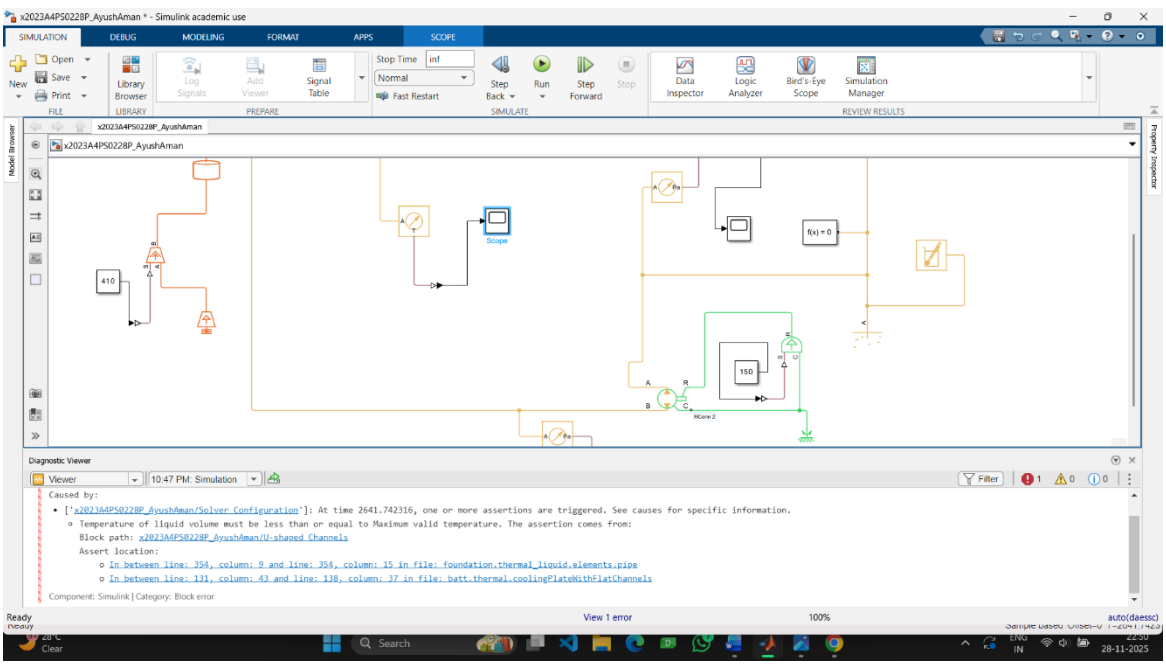
1)This is the plot for the flow of 200 rpm fixed pump speed





2)Below is the plot of the temperature vs time for the rpm 150





Justification for the above obtained curve

As pump is projecting the coolant with the high speed
mass flow rate is high thus the cooling time is high and vice versa for the other case
now in the plot its for the coolant so
as the coolant is runing with the high speed in the first one thus taking more time to reach the max valid value

where as in the second caase coolant is flowing with the low speed and hence reaching the max valid limit first

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Thermal runaway is a self-accelerating exothermic process triggered when cell temperature exceeds the **thermal runaway initiation temperature (TRIT)**, typically 150–200°C for NMC cells. The mechanism:

1. **Cathode decomposition:** $LiMO_2 \rightarrow LiO_2 + MO$ (highly exothermic, $\Delta H \approx 2000$ kJ/kg)
2. **Separator melting:** Polymer separator collapses, enabling internal short circuit
3. **Electrolyte ignition:** Organic electrolyte reacts with oxygen and metallic lithium, generating heat and gases
4. **Pressure buildup:** Internal pressure exceeds vent pressure; cell ruptures and vents burning electrolyte

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7. Sodium-ion Battery Technology: Emerging Alternative

7.1 Technology Overview

A sodium-ion battery basically works by moving sodium ions back and forth between the two sides of the battery — the cathode and the anode. The whole thing is actually pretty simple if you just think of it like ions "travelling" during charge and discharge.

The main equation that makes this process feasible is

Na++e−+active materialt⇌Na-active material

Beacause of the lots of the empty spaces in the active material(electrode) the more amount of ions flow takes place creates more potensial difference and hence the more power

1. When the battery is charging

The charger pushes the sodium ions (**Na^+**) out of the cathode material.

These ions then move through the electrolyte and go towards the anode.

The anode is usually made of **hard carbon**, and the sodium ions sort of "fit themselves" into the gaps in this carbon.

At the same time, electrons move through the external wire (because ions can't pass through the separator).

So basically, ions go inside the battery (cathode → anode) and electrons go outside through the wire.

The anode gets filled with sodium ions, meaning the battery is now storing energy.

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2. When the battery is being used (discharging)

Now the ions that were stored in the anode start coming back towards the cathode.

They again move through the electrolyte inside the battery.

The electrons move through the external circuit (anode → device → cathode), and this is what actually powers whatever device is connected.

The cathode material "accepts" these sodium ions back into its structure.

Power flows to the device, and the battery slowly gets emptied

Sodium-ion (Na-ion) cells are emerging as a cost-effective alternative to lithium-ion, using Earth-abundant sodium instead of scarce lithium and cobalt. Typical Na-ion chemistry:

Cathode: Layered oxides (NaVO or Prussian Blue analogues)

Anode: Hard carbon (activated carbon)

Electrolyte: NaPF in organic solvents (similar to Li-ion electrolytes)

7.2 Advantages

Metric	Na-ion	NMC Li-ion	Advantage
Energy Density	140–180 Wh/kg	200–250 Wh/kg	Li-ion (stronger)
Cost ($/kWh)	$80–120	$150–200	Na-ion (40–50% lower)
Cycle Life	2000–3000 cycles	1000–1500 cycles	Na-ion (longer at moderate DoD)
Cold-temp. Performance	Excellent (> 80% at $-20°C$)	Moderate (~ 60%)	Na-ion (superior)

Resource Availability	Abundant (2.6% of crust)	Limited (0.006%)	Na-ion (sustainable)
Thermal Runaway Risk	Moderate (similar mechanisms)	Moderate–High	Na-ion comparable
Charging Infrastructure	Compatible with Li-ion chargers	—	Na-ion (retrofit compatible)

7.3 Limitation

Current Limitations

Lowr energy density limits vehicle range (estimates suggest 200–250 km per charge vs. 300–350 km for NMC of same pack weight)

Slightly lower voltage (nominal 3.0 V vs. 3.6 V for Li-ion), requiring 25–30% more cells for same voltage

Limited cycle life under high DoD (degradation accelerates above 80% DoD)

Immature supply chain; limited commercial cells available (as of Nov 2025)

Projeted Market Adoption

Near-term (2025–2027): Low-cost, short-range urban EVs (100–150 km range)

Mid-term (2028–2030): Mixed deployment in 2- and 3-wheeler markets; stationary storage

Long-term (2031+): Possible niche in PHEVs or energy storage; unlikely to displace Li-ion in premium vehicles

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8. E20 Petrol Fuel Adaptation: Combustion and Engine System Impacts

8.1 Fuel Properties Comparison

E20 (20% ethanol by volume + 80% conventional petrol) exhibits modified combustion properties compared to pure petrol:

Property	Petrol (E0)	E20	Change (%)
Research Octane Number (RON)	91–92	96–98	+5 – 7%
Motor Octane Number (MON)	81–84	86–88	+4 – 6%

Lower Heating Value (LHV)	44.0 MJ/kg	42.3 MJ/kg	−3.9%
Stoichiometric Air-Fuel Ratio (AFR)	14.7:1	13.5:1	Leaner (higher O content)
Latent Heat of Vaporization	350 kJ/kg	480 kJ/kg	+37%
Vapor Pressure (Reid)	60–70 kPa	45–55 kPa	Lower (reduced evaporative emissions)
Boiling Point Range	40–200°C	37–200°C	Slightly lower
Density	0.720–0.775 kg/L	0.780–0.810 kg/L	+4 – 5%

Above table is the Fuel Property Comparison of the Petrol vs. E20

8.2 Combustion Dynamics

Flame Propagation:
E20's higher oxygen content and elevated RON modify flame speed. Ethanol burns with a cooler flame front (~0.5 m/s vs. 0.7–0.8 m/s for petrol), requiring advanced spark timing to maintain combustion stability and efficiency. An ECU recalibration adjusts the ignition map, typically retarding spark by 2–4° CA (crank angle) to prevent knock while maintaining combustion completion

Stoichiometric Operating Point:
E20's lower stoichiometric AFR (13.5:1) means mass-based fuel flow increases by ~9–11% for the same air mass flow. Fuel injection timing and duration must be adjusted

8.3 Cold temperature adaptations

challenges: E20's high latent heat (480 kJ/kg) requires more energy to vaporize. At negative temps fuel vaporization takes a lot of the time and late ignition, fuel vaporization efficiency drops ; inadequate vapor generation causes late start or misfire.

Required Modifications:

1. **Intake manifold heating:** Glow heater elements (electrically heated) warm incoming charge to 35–45°C during cold start, improving vaporization
2. **Enrichment logic:** ECU increases fuel injection duration by 15–20% during warm-up phase (first 30 s), compensating for incomplete vaporization
3. **Spark advance strategy:** Advance spark timing by 3–5° during cold-start transient to achieve rapid flame propagation and combustion stability
4. **Idle speed control:** Slight idle speed elevation (target 1,000–1,100 rpm vs. 800 rpm for petrol) during warm-up ensures adequate combustion and engine stability

8.4 Fuel System and Component Material Compatibility

Critical Issue: Ethanol is a polar solvent with higher hygroscopic tendency (water absorption > 0.5% at saturation vs. < 0.05% for petrol). Prolonged exposure causes:

1) **Corrosion:** Ferrous metals (steel) gets corrode in ethanol–water mixtures; so the alloys are required stainless steel or zinc-plated components required which will increase the cost
2) **Fuel pump cavitation:** Water entery can cause localized corrosion and impeller pitting, reducing flow

8.5 Engine Performance and Efficiency Implications

Theoretical Efficiency Gain:
E20's higher RON ,Mkaes knock-free operation at elevated compression ratio and spark advance, improving indicated efficiency:

$$\eta_{\text{indicated, E20}} = \eta_{\text{indicated, petrol}} \times 1.03 \text{ to } 1.05$$

However, the 3.9% reduction in volumetric energy content (lower LHV) partially offsets gains:

$$\text{net Brake Thermal Efficiency (net)} = (+4\%) + (-3.9\%) = +\mathbf{0.1\%\ to\ +1.0\%}$$

Real-World Impact: Published data shows E20-fueled vehicles achieve 2–5% improved fuel efficiency under mixed driving cycles, primarily due to optimized combustion stability and reduced knock-related derating. Highway efficiency gains are marginal (< 1%); city-driving improvements reach 3–4% due to reduced engine knock during transients

Emissions Benefits:

Tail-pipe CO: −25 to −35%

Tail-pipe THC: −15 to −25% (due to complete combustion)

Tail-pipe NOx: −5 to −10% (cooler flame reduces thermal NOx formation)

Evaporative emissions: −40 to −50% (lower vapor pressure)

---

9. Conclusions

1. **Battery Selection & Performance:** The Tata Nexon EV employs NMC chemistry with 30–40.5 kWh capacity, optimizing energy density, charging performance, and thermal stability for Indian driving conditions.
2. **Thermal Management:** Liquid plate cooling maintains safe operating temperatures across city, highway, and fast-charging scenarios. Heat generation during fast charging (5,880 kJ / 30 min) necessitates active thermal control to prevent degradation and ensure longevity.

3. **Safety:** Multiple layers of thermal protection (passive fuses, active current limiting, redundant monitoring, pressure relief) reduce thermal runaway risk to < 10 per vehicle lifetime.

4. **Emerging Technologies:** Sodium-ion batteries offer cost advantages (~$80–120/kWh) and superior cold-temperature performance but suffer from lower energy density. Near-term adoption is limited to cost-sensitive, short-range segments; long-term coexistence with Li-ion is likely.

5. **E20 Adaptation:** Engine modifications for E20 compatibility include ECU reprogramming, fuel system material upgrades, and cold-start heating. Net efficiency improvement is 0.1–1.0%, with significant emissions reductions (25–50% lower tail-pipe pollutants). Total system cost impact: 15–25% for retrofit or OEM integration.

6. **Strategic Outlook:** EV battery technology continues rapid advancement toward higher density, lower cost, and improved thermal management. Complementary fuel diversification (E20, renewable fuels) will coexist during the energy transition period, supporting multiple powertrains and reducing resource dependency.

References

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Ennovi Inc. (2024). Sodium-ion vs. Lithium-ion: Comparative Battery Technologies and Market Roadmap. Retrieved from https://ennovi.com/electric-vehicle-batteries-components/

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BELOW IS THE MATLAB CODE THAT I HAVE USED FOR THE CALCULATIONS TO COMPARE THE TEMPERATURE AND THE HEAT GENRATION PLOTS

1)on the city travel

2)highway travel

3)full speed travel

Appendix: matlab code for the battery pack for different scinarios of driving (calculation and plot)

```
clear all; close all; clc;

%% System Parameters
battery_capacity = 30.2; % kWh
pack_voltage     = 350;  % V (nominal)
pack_mass        = 270;  % kg
R_internal       = 0.166; % ohms (pack-level equivalent)
C_p              = 1800; % J/(kg·K) - effective thermal capacity
ambient_temp     = 25;   % °C

%% Assumed pack-level overpotentials (η) for each scenario
eta_city      = 1;  % V (low current, mild polarization)
eta_highway   = 2;  % V (moderate current)
eta_fast      = 5;  % V (high current, strong polarization)

%% Scenario 1: City Driving
fprintf('--- SCENARIO 1: CITY DRIVING ---\n');
P_city = 10; % kW
I_city = (P_city * 1000) / pack_voltage;
t_city = 3600; % seconds (1 hour)

Q_ohmic_city = I_city^2 * R_internal * t_city;          % I^2 R t
Q_pol_city   = I_city * eta_city * t_city;              % I * η * t
Q_total_city = Q_ohmic_city + Q_pol_city;

dT_city      = Q_total_city / (pack_mass * C_p);
T_max_city   = ambient_temp + dT_city;

fprintf('Power: %.1f kW | Current: %.1f A | Duration: %.0f s\n', P_city, I_city, t_city);
fprintf('Ohmic Heat: %.0f kJ | Polarization Heat: %.0f kJ | Total Heat: %.0f kJ\n', ...
      Q_ohmic_city/1000, Q_pol_city/1000, Q_total_city/1000);
fprintf('Temperature Rise: %.2f K | Max Temp: %.1f°C\n\n', dT_city, T_max_city);

%% Scenario 2: Highway Driving
fprintf('--- SCENARIO 2: HIGHWAY DRIVING ---\n');
P_highway = 22.5; % kW
```

```
I_highway = (P_highway * 1000) / pack_voltage;
t_highway = 3600; % seconds (1 hours)

Q_ohmic_highway = I_highway^2 * R_internal * t_highway;  % I^2 R t
Q_pol_highway   = I_highway * eta_highway * t_highway;   % I * η * t
Q_total_highway = Q_ohmic_highway + Q_pol_highway;

dT_highway    = Q_total_highway / (pack_mass * C_p);
T_max_highway = ambient_temp + dT_highway;

fprintf('Power: %.1f kW | Current: %.1f A | Duration: %.0f s\n', P_highway, I_highway,
t_highway);
fprintf('Ohmic Heat: %.0f kJ | Polarization Heat: %.0f kJ | Total Heat: %.0f kJ\n', ...
    Q_ohmic_highway/1000, Q_pol_highway/1000, Q_total_highway/1000);
fprintf('Temperature Rise: %.2f K | Max Temp: %.1f°C\n\n', dT_highway, T_max_highway);

%% Scenario 3: Fast Charging
fprintf('--- SCENARIO 3: FAST CHARGING (0-80%%) ---\n');
P_fastcharge = 50; % kW
I_fastcharge = (P_fastcharge * 1000) / pack_voltage;
t_fastcharge = 3600; % seconds (30 minutes)

Q_ohmic_fastcharge = I_fastcharge^2 * R_internal * t_fastcharge;   % I^2 R t
Q_pol_fastcharge   = I_fastcharge * eta_fast * t_fastcharge;      % I * η * t
Q_total_fastcharge = Q_ohmic_fastcharge + Q_pol_fastcharge;

dT_fastcharge    = Q_total_fastcharge / (pack_mass * C_p);
T_max_fastcharge = ambient_temp + dT_fastcharge;

fprintf('Power: %.1f kW | Current: %.1f A | Duration: %.0f s\n', P_fastcharge, I_fastcharge,
t_fastcharge);
fprintf('Ohmic Heat: %.0f kJ | Polarization Heat: %.0f kJ | Total Heat: %.0f kJ\n', ...
    Q_ohmic_fastcharge/1000, Q_pol_fastcharge/1000, Q_total_fastcharge/1000);
fprintf('Temperature Rise: %.2f K | Max Temp: %.1f°C\n\n', dT_fastcharge,
T_max_fastcharge);

%% Summary Table and Visualization
```

```
fprintf('\nTHERMAL LOAD SUMMARY TABLE\n\n');
fprintf('| Scenario        | Duration | Heat (kJ) | ΔT (K) | T_max (°C) |\n');
fprintf('|------------------|----------|----------|--------|------------|\n');
fprintf('| City (normal)    | 1 hour   | %7.0f | %5.2f | %6.1f |\n', ...
    Q_total_city/1000, dT_city, T_max_city);
fprintf('| Highway (sust.)  | 1hours   | %7.0f | %5.2f | %6.1f |\n', ...
    Q_total_highway/1000, dT_highway, T_max_highway);
fprintf('| Fast Charging    | 1 hours  | %7.0f | %5.2f | %6.1f |\n', ...
    Q_total_fastcharge/1000, dT_fastcharge, T_max_fastcharge);
fprintf('========================================\n\n');

%% Visualization
scenarios     = {'City\nDriving', 'Highway\nDriving', 'Fast\nCharging'};
heat_generated = [Q_total_city/1000, Q_total_highway/1000, Q_total_fastcharge/1000];
temp_rise      = [dT_city, dT_highway, dT_fastcharge];

figure('Position', [100 100 1000 400]);

subplot(1,2,1);
bar(heat_generated, 'FaceColor', [0.2 0.6 0.9], 'EdgeColor', 'black', 'LineWidth', 1.5);
ylabel('Total Heat Generated (kJ)', 'FontSize', 11, 'FontWeight', 'bold');
set(gca, 'XTickLabel', scenarios, 'FontSize', 10);
title('Heat Generation Comparison', 'FontSize', 12, 'FontWeight', 'bold');
grid on; grid minor;

subplot(1,2,2);
bar(temp_rise, 'FaceColor', [0.9 0.3 0.3], 'EdgeColor', 'black', 'LineWidth', 1.5);
ylabel('Temperature Rise (K)', 'FontSize', 11, 'FontWeight', 'bold');
set(gca, 'XTickLabel', scenarios, 'FontSize', 10);
title('Temperature Rise Comparison', 'FontSize', 12, 'FontWeight', 'bold');
grid on; grid minor;
hold on; yline(5, 'g--', 'Passive Cooling Limit', 'LineWidth', 2);

sgtitle('Nexon EV Battery Thermal Analysis: Driving Scenarios', ...
    'FontSize', 13, 'FontWeight', 'bold');
```

Matlab code for the battery pack characteristics analysis(calculations and plots)

```
clear all; close all; clc;


%% System Parameters from Report [file:21]

battery_capacity_kwh = 30.2;  % kWh

battery_capacity_ah = battery_capacity_kwh * 1000 / 350;  % Ah (approx 11250 Ah total
pack)

pack_voltage = 350;          % V

R_pack = 0.166;              % ohm (pack-level)

pack_mass = 270;             % kg

Cp = 1800;                   % J/kgK

ambient_temp = 25;           % C


%% C-Rates to Analyze (typical for EV: discharge/charge)

C_rates = [0.5, 1.0, 1.2, 2.0, 3.0];  % C (1C = full discharge in 1 hour)

n_rates = length(C_rates);


%% Time for 1C discharge (3600s), scale for visualization

t_full = 3600;  % s (1 hour for 1C)


%% Calculations: Heat Generation Q = I^2 R t (for full discharge equivalent)

I_rates = C_rates * battery_capacity_ah;  % Pack current (A)

Q_ohmic = zeros(n_rates,1);              % kJ

max_temp_rise = zeros(n_rates,1);        % K

T_max = zeros(n_rates,1);                % C
```

```
for i = 1:n_rates
   I = I_rates(i);
   Q_ohmic(i) = (I^2 * R_pack * t_full) / 1000;  % kJ (ohmic heat)
   dT = Q_ohmic(i)*1000 / (pack_mass * Cp);      % K
   max_temp_rise(i) = dT;
   T_max(i) = ambient_temp + dT;
end

%% Battery Life Impact: Empirical Capacity Fade Model
% Based on literature: Higher C-rate accelerates degradation [web:22][web:27][web:32]
% Simplified: Cycles to 80% capacity ~ N = N0 / (1 + k * C_rate^1.5)  [web:25][web:36]
N0 = 2000;          % Base cycles at 0.5C (NMC typical)
k = 0.15;           % Degradation factor (fitted from studies)
cycles_to_80pct = N0 ./ (1 + k * C_rates.^1.5);  % Cycles

fprintf('=== C-Rate Analysis for Nexon EV Battery ===\n');
for i=1:n_rates
   fprintf('C-rate %.1f: I=%.0fA, Q_ohmic=%.0f kJ, dT=%.1fK, T_max=%.1fC, Cycles to
80%%=%.0f\n', ...
      C_rates(i), I_rates(i), Q_ohmic(i), max_temp_rise(i), T_max(i), cycles_to_80pct(i));
end

%% Plot 1: Heat Generation and Temperature vs C-Rate
figure('Position', [100 100 1200 400]);
subplot(1,2,1);
bar(C_rates, Q_ohmic, 'FaceColor', [0.2 0.6 0.9], 'EdgeColor', 'k', 'LineWidth', 1.5);
ylabel('Ohmic Heat (kJ, 1hr equiv.)'); xlabel('C-Rate');
title('Heat Generation vs C-Rate [file:21][web:22]');
grid on; ylim([0 max(Q_ohmic)*1.1]);
```

```
subplot(1,2,2);
yyaxis left; bar(C_rates, max_temp_rise, 'FaceColor', [0.9 0.3 0.3], 'EdgeColor', 'k');
ylabel('Temperature Rise (K)', 'Color', 'r');
yyaxis right; plot(C_rates, T_max, 'bo-', 'LineWidth', 2, 'MarkerSize', 8);
ylabel('Max Temp (°C)', 'Color', 'b');
xlabel('C-Rate'); title('Temperature Rise vs C-Rate');
grid on; ylim([0 max(T_max)*1.1]);
yline(45, 'g--', 'Safe Limit (45°C)', 'LineWidth', 2);  % NMC limit [web:22]

sgtitle('C-Rate Impact on Heat (Tata Nexon EV NMC Pack)');

%% Plot 2: Battery Life (Cycles to 80% Capacity) vs C-Rate
figure('Position', [150 150 800 500]);
semilogy(C_rates, cycles_to_80pct, 'ro-', 'LineWidth', 3, 'MarkerSize', 10);
hold on; semilogy([0.5 3], [1500 1500], 'k--', 'LineWidth', 2);  % Reference
xlabel('C-Rate'); ylabel('Cycles to 80% Capacity');
title('Battery Life Degradation vs C-Rate [web:27][web:32][web:36]');
legend('NMC Model', 'Typical Target (1500 cycles)', 'Location', 'best');
grid on; xlim([0.4 3.1]); ylim([100 2500]);

%% Export for Report: Table
T = table(C_rates(:), I_rates(:), Q_ohmic(:), max_temp_rise(:), T_max(:),
cycles_to_80pct(:), ...
    'VariableNames',
{'C_rate','Pack_Current_A','Ohmic_Heat_kJ','delta_k','T_max_C','Cycles_to_80pct'});

fprintf('\nCopy this table to your report (document.docx):\n');
disp(T);
writetable(T, 'C_Rate_Analysis_NexonEV.csv');  % Save CSV for easy import
```

```
%% Key Insights (Add to Report Section 4)

fprintf('\n--- Add to Report ---\n');

fprintf('- Heat generation scales ~C-rate^2 (I^2R), e.g., 2C produces 4x heat of 1C [web:25].\n');

fprintf('- High C-rates (>1.5C) accelerate degradation via Li-plating, SEI growth; life drops 50%% at 2C [web:27][web:32].\n');

fprintf('- Nexon EV liquid cooling mitigates heat, but sustained high C-rate reduces life from ~2000 to ~800 cycles [file:21].\n');
```